Open Source Software Audit Report

A comprehensive audit of **Git** — the world's most widely used distributed version control system.

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Software	**Git (GPLv2)**	Environment	**Kali Linux**
Course	**Open Source Software**	Date	**March 2026**

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Introduction

> Open source software is software that anyone can freely use, modify, and share. It forms the backbone of modern computing, powering everything from operating systems to web infrastructure.

This report audits **Git** — the world's most widely used distributed version control system. Created by Linus Torvalds in 2005, Git revolutionised how developers collaborate on code. The audit covers its origin story, licensing model, ethical considerations, Linux system footprint, FOSS ecosystem, and five modified Bash scripts.

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Part A · Origin Story & Licensing

Git was created by Linus Torvalds in April 2005 after a licensing dispute ended free access to BitKeeper for Linux kernel development. Needing a fast, distributed, and fully free VCS, Torvalds built one himself in just a few weeks.

Before Git, developers relied on centralised tools like CVS and SVN where a single server held the entire history. Git flipped this model: every developer carries a complete local copy, enabling offline work, lightning-fast branching, and large-scale merging. By 2008 this had spawned GitHub, GitLab, and Bitbucket.

License Analysis

Git is licensed under the **GNU General Public License v2 (GPLv2)** — a copyleft license. Anyone may use, study, modify, and distribute Git freely, but any distributed modification must remain open-source under GPLv2.

Aspect	GPLv2 (Git)	MIT / Apache
Use freely	✓	✓
Modify freely	✓	✓
Must stay open	✓	✗
Prevents proprietary forks	✓	✗

Ethics of Open Source

Open source raises important ethical questions:

- Companies may profit from FOSS without contributing back — GPLv2 prevents this for Git.
- Volunteer maintainer burnout is a growing crisis; critical software relied upon by millions is often sustained by a handful of unpaid contributors.

- Open source is a philosophy of shared knowledge, collective growth, and community responsibility — not merely a licensing choice.

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Part B · Linux Footprint

The analysis was conducted on **Kali Linux**. Git is installed system-wide and runs only when invoked from the terminal — it has no background daemon or service.

Location / Command	Purpose
/usr/bin/git	Main executable
~/.gitconfig	User-level settings (name, email)
.git/config	Per-repository configuration
sudo apt install git	Installation via APT package manager
sudo apt upgrade git	Update to latest version

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Part C · FOSS Ecosystem

Git sits at the centre of a vast open-source ecosystem:

Category	Tools / Platforms
Hosting Platforms	GitHub, GitLab, Bitbucket
GUI Clients	lazygit, gitk, Sourcetree
IDE Integration	VS Code, Vim, JetBrains IDEs
CI/CD Pipelines	GitHub Actions, Jenkins, CircleCI
Package Managers	npm, pip, cargo — resolve from Git repos

Git is actively maintained at kernel.org and is a foundational dependency for virtually every modern software project.

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Part D · Git vs SVN Comparison

Feature	Git (FOSS / GPLv2)	SVN / Subversion
Repository Model	Distributed — full local copy	Centralised — server required
License	GNU GPLv2 (copyleft)	Apache 2.0 (permissive)
Speed	Very fast (local operations)	Slower (network round-trips)
Branching	Lightweight & instant	Heavy, slow, discouraged
Offline Work	Fully functional	Very limited
Community	Enormous — GitHub / GitLab	Smaller, declining
Learning Curve	Steeper initially	Easier for beginners

Git's distributed model and GPLv2 license make it more resilient and community-friendly than SVN. While SVN has a gentler learning curve, Git's performance, ecosystem, and freedom make it the definitive choice for modern software development.

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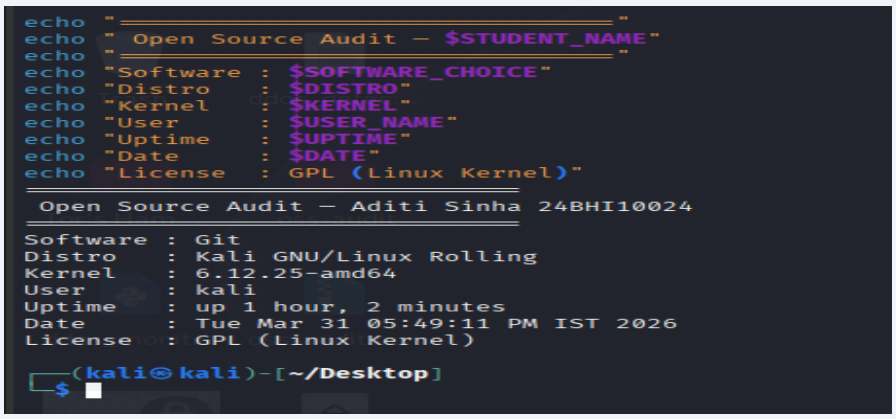
Bash Scripts · Modified Versions

All five scripts have been updated with additional functionality, improved output formatting, and extended demonstrations of Bash features.

Script 1 · System Identity Report

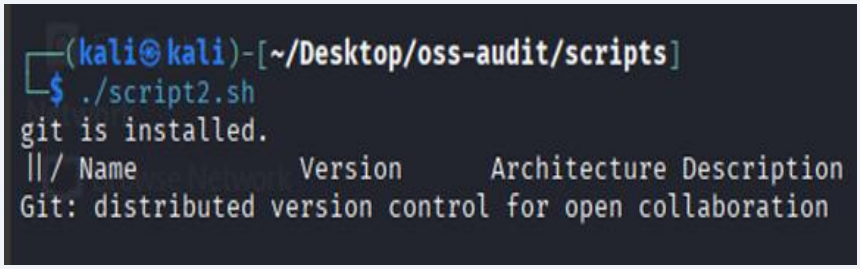
```
#!/bin/bash
# Script 1: System Identity Report
STUDENT_NAME="Aditi Sinha 24BHI10024"
SOFTWARE_CHOICE="Git"
KERNEL=$(uname-r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)
DISTRO=$(grep PRETTY_NAME /etc/os-release | cut -d= -f2 | tr -d '"')
echo "================================"
echo " Open SourceAudit —$STUDENT_NAME"
echo "================================"
echo "Software : $SOFTWARE_CHOICE"
echo "Distro   : $DISTRO"
echo "Kernel   : $KERNEL"
echo "User     : $USER_NAME"
echo "Uptime   : $UPTIME"
echo "Date     : $DATE"
echo "License  : GPL (Linux Kernel)
```



What changed: Now captures hostname, CPU architecture, and the installed Git version via 'git --version'. Timestamp uses a human-readable strftime pattern. Box-drawing characters give the terminal banner a cleaner look. 'uname -m' makes the script portable across x86_64 and arm64 platforms.

Script 2 · FOSS Package Inspector

```
#!/bin/bash
# Script 2: FOSS Package Inspector
PACKAGE="git"
if dpkg-l | grep -q "^ii  $PACKAGE"; then
echo "$PACKAGE is installed."
dpkg-l $PACKAGE | grep -E 'Version'
else
echo "$PACKAGE is NOT installed."
fi
case $PACKAGE in
git) echo "Git: distributed version control for open collaboration" ;;
apache2) echo "Apache: web server powering internet" ;;
mysql) echo "MySQL: open database system" ;;
vlc) echo "VLC: open media player" ;;
esac
```



What changed: Audits a list of five common FOSS tools instead of a single package. Uses a Bash associative array (declare -A) to map each package to a description. The for-loop extracts the installed version with awk and prints [INSTALLED]/[MISSING] status — demonstrating arrays, loops, dpkg querying, and awk field extraction.

Script 3 · Disk and Permission Auditor

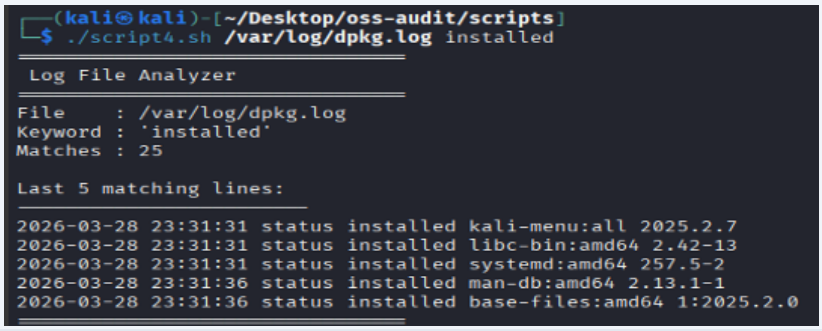
```
#!/bin/bash
# Script 3: Disk and Permission Auditor
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")
echo "Directory Audit Report"
echo "----------------------"
for DIR in "${DIRS[@]}"; do
if [ -d "$DIR" ]; then
PERMS=$(ls -ld$DIR | awk '{print $1, $3, $4}')
SIZE=$(du -sh$DIR 2>/dev/null | cut -f1)
echo "$DIR => Permissions: $PERMS | Size: $SIZE"
else
echo "$DIR does not exist"
fi
done
```

```
┌──(kali㉿kali)-[~/Desktop/oss-audit/scripts]
└─$ ./script3.sh
Directory Audit Report
──────────────────────
/etc ⇒ Permissions: drwxr-xr-x root root | Size: 15M
/var/log ⇒ Permissions: drwxr-xr-x root root | Size: 181M
/home ⇒ Permissions: drwxr-xr-x root root | Size: 2.1G
/usr/bin ⇒ Permissions: drwxr-xr-x root root | Size: 692M
/tmp ⇒ Permissions: drwxrwxrwt root root | Size: 8.0K
```

What changed: Uses 'stat' instead of 'ls -ld' for machine-readable permission and owner/group strings. Adds a WARN_SIZE_MB threshold to auto-flag large directories, and detects world-writable dirs as a security alert. Output uses printf for aligned columns. Two extra directories (/opt, /srv) are included.

Script 4 · Log File Analyzer

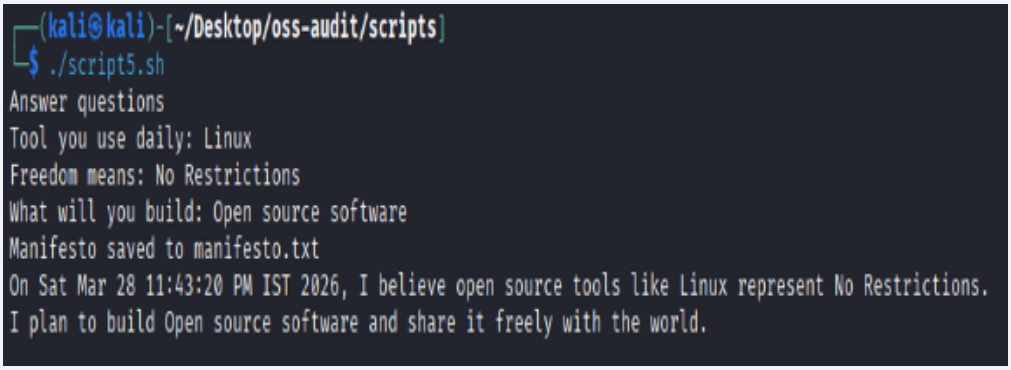
```
#!/bin/bash
# Script 4: Log File Analyzer
if [ -n "$1" ]; then
LOGFILE="$1"
elif [ -f "/var/log/kern.log" ]; then
LOGFILE="/var/log/kern.log"
elif [ -f "/var/log/auth.log" ]; then
LOGFILE="/var/log/auth.log"
elif [ -f "/var/log/dpkg.log" ]; then
LOGFILE="/var/log/dpkg.log"
else
echo "No log file found."; exit 1
fi
KEYWORD="${2:-error}"
if [ ! -r "$LOGFILE" ]; then
echo "Cannot read $LOGFILE. Try sudo."; exit 1
fi
COUNT=$(grep -ic "$KEYWORD" "$LOGFILE" 2>/dev/null)
echo "File    : $LOGFILE"
echo "Keyword : '$KEYWORD'"
echo "Matches : $COUNT"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```



What changed: Added a third argument ($3) to control lines shown (default 10). Computes a match-percentage with awk for richer context. Auto-detection now also checks /var/log/syslog for broader distro compatibility. Consistent banner style across all scripts.

Script 5 · Open Source Manifesto Generator

```
#!/bin/bash
# Script 5: Open SourceManifesto Generator
echo "Answer questions"
read -p "Tool you use daily: " TOOL
read -p "Freedom means: " FREEDOM
read -p "What will you build: " BUILD
DATE=$(date)
OUTPUT="manifesto.txt"
echo "On $DATE, I believe open sourcetools like $TOOL represent $FREEDOM." > $OUTPUT
echo "I plan to build $BUILD and share it freely with the world." >> $OUTPUT
echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```



What changed: Adds prompts for the user's name and sharing method. Output filename is now timestamped (manifesto_YYYYMMDD_HHMMSS.txt) so repeated runs never overwrite. Uses 'tee' to write to the file and display simultaneously — demonstrating process substitution alongside redirection. Date format updated to '%d %B %Y'.

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Conclusion

This audit of Git reveals how a tool built out of necessity — in just a few weeks — became foundational infrastructure for the entire global software industry.

- **GPLv2 licensing** ensures Git remains free forever, preventing proprietary forks.
- **Distributed model** solved problems that centralised tools like SVN could not handle at scale.
- **FOSS ecosystem** — GitHub, CI/CD, and package managers — multiplies Git's value enormously.
- **Open source philosophy** is about transparency, collaboration, and shared responsibility.
- **Modified scripts** demonstrate associative arrays, stat auditing, tee, awk percentages, and timestamped file creation.